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BT 501 Current Final Terms Spring 2020

2 Marks

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Qno1: Define Mitochondrial Homeostasis.

Because neurons require considerable energy to maintain their metabolically active state, they rely heavily on the maintenance of mitochondria in a functional state, a property known as mitochondrial homeostasis.

Qno2: Write down four important function of Immune system.

1. Immune reorganization

2. Contain and eliminate infection

3. Immune regulation

4. Immunological memory

Qno3: Write down the modifications which are also there for creating transgenic.

A few modifications are also there for creating transgenic:

Electroporation of DNA into the sperm, ova, orθ embryos.

Nuclear transfer with somatic or embryonic cells.

Qno4: Why CA125 test is not recommended as a screening test of ovarian cancers?

CA125 test is not recommended as a screening test of ovarian cancers, because CA125-a high molecular-weight glycoprotein

• Tumor Marker Non specific –increase in other tumors/noncancerous disorders Measuring Disease-Associated Proteins by Sandwich ELISA

• Research- other biomarkers -more specific for ovarian cancer.

Qno5: Define passive immunity.

Development of immunity between the mother and child is called passive immunity because the fetus does not make any antibodies on its own; it only borrows them from the mother.

Qno6: Write about the somatic cell nuclear transfer technique.

• Due to the absence of proven ES cells and the recent advances in nuclear transfer (NT), current emphasis for creating transgenic animals is on somatic cell nuclear transfer (SCNT).

• Nuclear transfer is a technique that is used to create a genetically identical copy, or a clone, of an animal.

• Nuclear transfer commonly involves the transfer or placement of a donor nucleus into the cytoplasm of an enucleated MII oocyte.

Qno7: Define Autologous cells.

The autologous cell transplantation is basically a transplantation in which cells obtained from patients can be transplanted into the same patient.

Qno8: Write down two advantages of Live attenuated vaccine.

The advantages of attenuated vaccines Single dose is effective, and boosters are not

♣ Essentially required because immune system is exposed to the live organism for prolonged period.

♣Efficiently activates adaptive immune responses.

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3 Marks

Qno1: Write about the Hemagglutination test to determine blood type.

Hemagglutination test to determine blood type.

(A) Antiserum containing antibodies against A or B surface antigens is mixed with red blood cells carrying A surface antigens. Clumping (agglutination) in the presence of anti-A antiserum but not with anti-B antiserum indicates blood type A.

(B) Agglutination responses for different blood types.

Qno2: How many types of immune disease. Write the name.

There are three types of immune disease.

a) Immunodeficiency

b) Autoimmune disease

c) Hypersensitivity

Qno3: Write down a note on Recombinant blood clotting factor VIII.

Factor VIII is one of the important factors of all blood clotting factors. Deficiency of factor VIII causes bleeding disorder called hemophilia A. The hemophilia may be mild to severe depending upon factor VIII concentration in the body.

In moderate and severe factor VIII deficiency, there can be spontaneous bleeding episodes in the joints. Hemophilia A affects 1 in 5,000–10,000 males. Replacement therapy is the treatment option for hemophiliacs either with human plasma-derived factor VIII (pdFVIII) or recombinant FVIII (rFVIII).

Transfection of HEK 293 cell cultures in serum-free suspension is being tried for optimal yield. Recombinant factor VIII (rFVIII) is produced by culturing mammalian cells as baby hamster kidney (BHK) or Chinese hamster ovary cells (CHO), using large-scale bioreactors . Standardizations are done to maximize yields.

Qno4: Write the three trisomy with disorder name and number of chromosome.

Trisomy 13, the presence of three copies of chromosome 13, causes Patau syndrome.

Trisomy 18, the presence of 3 copies of chromosome 18, elicits Edward syndrome.

Trisomy 21, the first human chromosomal disorder discovered (in 1959), is an abnormality that displays an extra copy (total of 3 copies) of chromosome 21 and causes Down syndrome.

Qno5: Define immunodeficiency and its types.

Immunodeficiency is a physiological condition where the immune system's capacity to fight infectious disease has become dysfunctional or entirely absent.

Primary immunodeficiency

A number of rare diseases have been reported susceptible to infections from childhood onward and many of these diseases are hereditary and autosomal recessive or X-linked in nature. There are more than 80 primary immunodeficiency syndromes reported in humans and they are generally grouped by their malfunctioning of lymphocytes or granulocyte cells. The treatment of primary immunodeficiencies depends on the nature of the defect, and may involve antibody infusions, long-term antibiotics, and stem cell transplantation.

Acquired immunodeficiency

An immune deficiency that is caused due to external processes is called secondary or acquired immunodeficiency. The common causes for secondary immunodeficiency are malnutrition, aging, and particular medications. It has been reported that there are many specific diseases that directly or indirectly impair the immune system Immunodeficiency is also the hallmark of AIDS, caused by HIV.

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Qno6: Write down two basic strategies for producing transgenic animals.

There are two basic strategies for producing transgenic animals.

• These are the so-called “gain of function” or “loss of function” transgenics.

• The “gain of function” relies on addition of a cloned fragment of DNA to an animal’s genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

Qno7: Write hybridization probe which one of DNA sequence is specific diagnostic approach.

Hybridization is the formation of hydrogen bonds between two complementary strands of nucleic acids. A diagnostic test involving DNA hybridization utilizes a DNA probe to detect a complementary target DNA sequence that is characteristic of the disease. The probe is labeled with a reporter molecule that indicates hybridization between the target and probe DNA. Hybridization probes are often employed to detect the presence of microbial pathogens. Malaria, caused by the parasite Plasmodium falciparum, is one of the most common infectious diseases and is especially fatal in young children.

Qno8: Define and classify the immunological memory.

When B cells and T cells are activated and begin to reproduce, some of their offspring can become long lived memory cells. This process is called immunological memory.

Immunological memory can be classified as passive short-term memory or active long-term memory.

5 Marks

Qno1: Write a note on Fragile X syndrome.

• Fragile X syndrome is caused by a —fragile‖ site at the end of the long arm of the X chromosome.

• Fragile X syndrome is the most common cause of inherited mental retardation.

• Although it is an X-chromosome-linked recessive trait with variable expression and incomplete penetrance, 30% of all carrier women are affected.

• Fragile X syndrome is caused by loss-of-function mutations in the fragile X mental retardation 1 (FMR1) gene.

• FMR1 encodes the FMRP protein found in many tissues and at particularly high levels in the brain and testes

• The FMRP protein may help regulate synaptic plasticity and thereby control learning and memory.

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• Fragile X syndrome belongs to a growing class of neurodegenerative disorders known as trinucleotide repeat disorders. • Among these disorders, 14 affect humans and elicit neurological dysfunction.

• Trinucleotide CGG repeat expansions (200 to more than 1,000 repeats) that inactivate the FMR1 gene are the most common mutations observed at this locus.

Qno2: Write down the advantages of PCR based test.

Advantages: specificity, sensitivity, amplification (subnational amount of a target sequence), multiplexing, low cost.

Qno3: Write down the advantages of DNA Vaccine.

The advantages

• Protein is expressed in its native conformation.

• Long-term exposure of immune system to the antigen.

• Efficient activation of cell-mediated and humoral immune systems leading to memory response.

• There is no requirement of specific storage condition like refrigeration of plasmid DNA, and it reduces the cost and is easy to handle.

• The genes expressing cytokines can be ligated to the vector, and the cytokine produced at the site may activate the desirable branch of immune system.

• Vaccine is safe, without any infectious agent.

• Easier to produce in large quantity.

Qno4: Write a note on Genetic Tay Sach disease.

• Tay–Sachs disease is an autosomal recessive fatal genetic disorder caused by a genetic mutation in the HEXA gene on human chromosome 15, which encodes the hexosaminidase A enzyme.

• This mutation leads to a decrease in function of hexosaminidase A, and as a result, harmful quantities of a sphingoglycolipid, termed ganglioside GM2, accumulate in brain neurons

• In the most common form of the disease (infantile Tay–Sachs), abnormal hexosaminidase A enzyme activity and the accompanying harmful accumulation of cell membrane associated gangliosides in neurons lead to premature neuronal death, paralysis, dementia, blindness, psychoses, and even death of the patient.

Tay–Sachs disease may be diagnosed by a blood test that measures levels of the hexosaminidase A enzyme in serum, lymphocytes, or skin fibroblasts

• Prenatal tests of hexosaminidase A activity, such as amniocentesis (at 15 to 16 weeks of pregnancy) and chorionic villus sampling (at 10 to 12 weeks of pregnancy), can now diagnose Tay–Sachs disease in the fetus.

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Qno5: Write a note Recombinant Human Insulin.

• Insulin is a peptide hormone consisting of 51 amino acids. It is secreted by β cells of islets of Langerhans of pancreas.

• The hormone is responsible for maintaining normal blood glucose level in blood.

• Insulin is stored in the form of proinsulin which contains two polypeptide chains, A and B, and is connected with a third peptide C chain, which before secretion is cleaved with production of insulin and C-chain.

• The cleavage results in the removal of C-chain, and the A (21 amino acids) and B chain (30 amino acids) are linked by disulfide linkage to form mature insulin

• In the beginning, efforts were made to isolate mRNA for pre- and proinsulin from rat islets of Langerhans of pancreas and to synthesize cDNA.

• Thereafter, it was inserted into a plasmid. The recombinant plasmids were transferred into the E. coli cells, which secreted proinsulin.

• Scientists have chemically synthesized DNA sequences for two chains, A and B, of insulin and separately inserted into two pBR322 plasmids by the side of β-galactosidase gene.

Qno6: Define rDNA technology and write its step of gene cloning in rDNA technology.

Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin. The inserted DNA/gene is known as transgene. Transgene is introduced into the organism, using recombinant DNA technology.

Gene cloning has four stages:

(1) Generation of DNA fragments,

(2) Ligation to a vector,

(3) Propagation in a host cell,

(4) Selection of the transformed host.

Qno7: How can two different alleles are distinguish by using a Taq man Assay.

To simultaneously detect normal and disease-associated alleles, two TaqMan probes are utilized. Each probe is exactly complementary to either the normal or the disease-associated DNA sequence, and each probe has a different fluorescent dye attached to its 5′ end (Fig.). Intact probes, whether unbound or bound to template DNA, do not fluoresce because of the presence of a quencher molecule at the 3′ end of the probe. PCR primers anneal to sequences that flank the probe hybridization site, and as PCR amplification proceeds, the TaqMan

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probe is displaced by the growing DNA strand. The 5′ nuclease activity of the Taq polymerase degrades the 5′ end of the TaqMan probe, thereby releasing the fluorescent dye and removing it from the proximity of the quencher molecule. Thus, only TaqMan probes that were previously bound to target DNA are degraded and subsequently fluoresce. Any mismatched probes that are not complementary to the SNP in the region where the TaqMan probe binds will be displaced but not cleaved, so they will not fluoresce. By monitoring the fluorescence at two different wavelengths (one for each TaqMan probe), it is possible to **distinguish** individuals that are homozygous for each SNP (either normal or disease associated) and heterozygous (carrying one normal and one disease-associated allele). This technique may be used to detect more than two different SNPs at the same time as long as the fluorescent dyes attached to each SNP-specific probe have well-separated, non-overlapping fluorescence maxima.

10 Marks

Qno1: Write a note on embryonic stem cell mediated gene-transfer method for producing transgenic animals.

• Embryonic stem cells are obtained from inner cell mass of blastocyst.

• This method involves prior insertion of the desired DNA sequence by homologous recombination into an in vitro culture of embryonic stem (ES) cells, and the transformed cells can be selected using standard markers.

• Cells containing the desired DNA are incorporated into the blastocoel of host's embryo at the blastocyst stage, where they mix with inner cell mass

• Unlike the other two methods, which require live transgenic offspring to test for the presence of the desired transgene.

• This method allows testing for transgene at the cell stage.

• This is advantageous as there is no convenient way to select for eggs or embryos that have taken up foreign DNA.

• ES cell-mediated gene transfer is the method of choice for gene inactivation, the so-called knockout method.

• This technique is of particular importance for the study of the genetic control of developmental processes.

• This technique works particularly well in mice. It has the advantage of allowing precise targeting of defined mutations in the gene via homologous recombination.

Qno2: Define rDNA technology and write its step of gene cloning in rDNA technology.

V. IMPORTANT

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Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin. The inserted DNA/gene is known as transgene. Transgene is introduced into the organism, using recombinant DNA technology.

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• The “gain of function” relies on addition of a cloned fragment of DNA to an animal’s genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

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